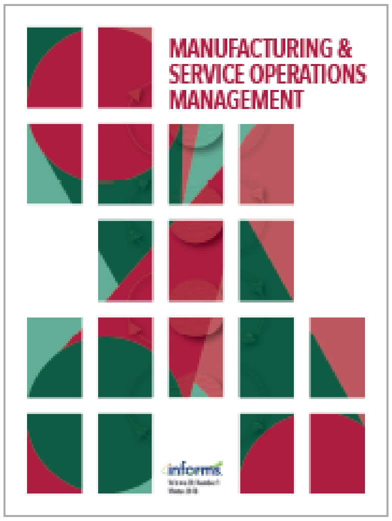


Manufacturing & Service Operations Management

Publication details, including instructions for authors and subscription information:
http://pubsonline.informs.org

Value of Online—Off-line Return Partnership to Off-line Retailers

Elina H. Hwang, Leela Nageswaran, Soo-Haeng Cho

informs
http://pubsonline.informs.org/journal/msom

MANUFACTURING & SERVICE OPERATIONS MANAGEMENT
Vol. 24, No. 3, May–June 2022, pp. 1630–1649
ISSN 1523-4614 (print), ISSN 1526-5498 (online)

Value of Online–Off-line Return Partnership to Off-line Retailers

Elina H. Hwang,[a] Leela Nageswaran,[a] Soo-Haeng Cho[b]

[a] Michael G. Foster School of Business, University of Washington, Seattle, Washington 98195; [b] Tepper School of Business, Carnegie Mellon University, Pittsburgh, Pennsylvania 15213

Contact: elina7@uw.edu, https://orcid.org/0000-0002-8717-7508 (EHH); lnages@uw.edu, https://orcid.org/0000-0003-3649-1174 (LN); soohaeng@andrew.cmu.edu, https://orcid.org/0000-0001-5158-7591 (S-HC)

Received: May 7, 2020
Revised: March 17, 2021; May 19, 2021; June 13, 2021
Accepted: June 14, 2021
Published Online in Articles in Advance: November 16, 2021

https://doi.org/10.1287/msom.2021.1026

Abstract. ***Problem definition***: This paper examines whether and, if so, how much an online–off-line return partnership between online and third-party retailers with physical stores (or "location partners") generates additional value to location partners. ***Academic/practical relevance***: Online shoppers often prefer to return products to stores rather than mailing them back. Many online retailers have recently started to collaborate with location partners to offer the store return option to their customers, and we quantify its economic benefit to a location partner. ***Methodology***: We analyze proprietary data sets from Happy Returns (which provides return services for more than 30 online retailers) and one of its location partners, using a panel difference-in-differences model. In our study, a treatment is the initiation of the return service at each of the location partner's stores, and an outcome is the store and online channel performance of the location partner. We then explore the mechanisms of underlying customer behavior that drive these outcomes. ***Results***: We find that the partnership increases the number of unique customers, items sold, and net revenue in both store and online channels. We identify two drivers for this improved performance: (1) the location partner acquires new customers in both store and online channels, and (2) existing customers change their shopping patterns only in the store channel after using the return service; in particular, they visit stores more often, purchase more items, and generate higher revenue after their first return service. ***Managerial implications***: To our knowledge, we provide the first direct empirical evidence of value to location partners from a return partnership, and as these partnerships become more prevalent, our findings have important managerial implications for location partners and online retailers alike.

Supplemental Material: The online appendix is available at https://doi.org/10.1287/msom.2021.1026.

Keywords: econometric analysis • retailing • empirical research • service operations

1. Introduction

E-commerce (online) sales continue to grow. In the United States, e-commerce accounts for 21.2% of total retail sales, excluding autos and gasoline, in the fourth quarter 2020, jumping from 18.0% and 16.4% in the fourth quarter 2019 and 2018, respectively (Kapner 2021). In this era of online shopping, off-line stores are struggling to survive: in 2019, U.S. retailers announced 9,302 store closings, a 59% jump from 2018 (Valinsky 2020). A recent study predicted that another 75,000 U.S. stores would be forced out of business by 2026 (Danziger 2019). Behind the growth of online shopping is an inevitable side effect: more items are getting returned. About 5% to 10% of in-store purchases are returned, but that rises to 15% to 40% for online purchases (Reagan 2019).

As customers express that a return process is the least favorite part of online shopping (Mulpuru 2017), online retailers seek ways to make their return process more convenient for customers. In a recent consumer survey, for retailers with a physical presence, 46.7% of shoppers prefer to return products to stores, and just 26.7% prefer mailing them back (Risley 2018). This is because mailing back returns can be a hassle for consumers: they have to repackage an item, print a return label if it is not included in the original box, get it to a carrier, and wait for the refund into their account. There may also be a return fee for returns via mail, which is often waived for returns to stores. Interestingly, consumers find drop-off locations, such as a convenience store, even more convenient because of the possibility to shop for other items at the store (Narvar 2019). Unfortunately, online retailers do not have off-line stores to offer the preferred store return option for their consumers. (Throughout this paper, "online retailers" refers to online retailers with no physical stores as well as omnichannel retailers with a limited number of physical stores, for which most consumers do not have a store return option.)

To address such a fundamental shortcoming of the online business model, online retailers have started to collaborate with third-party retailers with physical stores. Amazon customers can now return eligible items to any of more than 1,150 Kohl's stores for free (Kohl's 2019). Happy Returns is an intermediary company that provides return and logistics solutions for online retailers through its so-called "return bars" in more than 700 stores of its location partners (e.g., Cost Plus World Market, Office Depot, various shopping malls) (Happy Returns 2020). More than 30 online retailers, such as Everlane and Rothy's, use services from Happy Returns to allow online purchases to be returned in person to off-line stores. For store returns to Kohl's and Happy Returns' location partners, consumers can bring in items purchased online without repackaging them or printing a return label and get an immediate refund. FedEx has also started to offer return services to online retailers, and it partners with off-line stores, such as Walgreens and Dollar General, to expand its drop-off locations for returns (Forde 2019).

This paper studies this recent trend of partnerships between online and third-party retailers with physical stores for product returns with a focus on the economic benefits of this partnership to the third-party retailers. For convenience, in the rest of this paper, we call these partnerships "return partnerships," third-party (possibly omnichannel) retailers with physical stores "location (retail) partners," physical stores "off-line stores," in-person store returns "off-line returns," and off-line return service "return service." On the one hand, it is plausible that the economic benefit of a return partnership is negligible: this occurs if online retailers cater to a customer demographic that has limited overlap with the location partner, in which case an increase in adoption of the return service does not translate into economic value to the location partner or if there is a change in customers' purchasing channel choice (i.e., a *channel shift*) so that any improvement in store performance is due to reduced online sales of the location partner.

On the other hand, there may be a positive benefit to the location partner, in which case there may be two potential mechanisms that drive economic value. The first mechanism is a *customer acquisition effect*, which arises when the presence of the new return option brings in new customers who become familiar with the location partner's brand, and subsequently, these customers purchase the location partner's products. These purchases not only include the purchases made by store customers either during their return trips or in the future on nonreturn days, but also include the purchases made by online customers who become acquainted with the location partner through the return service offered to other online retailers. The second mechanism is a *customer supercharging effect*, which arises when existing customers of the location partner purchase more and/or more frequently with the location partner as a result of their adoption of the new channel of interaction. Whereas a similar effect was observed when customers visit an omnichannel firm's own physical showrooms in Bell et al. (2020), the effect in our setting comes from customers' use of a new service offered by the firm for a different retailer and has not been explored previously. We rigorously test for these mechanisms in our study.

Preliminary industry analysis suggests that return partnerships may be promising to location partners: foot traffic to Kohl's stores increased 24% in the three weeks after the Amazon returns program launched compared with the previous three weeks—in particular, visits lasting more than 16 minutes, which indicate a potential sales lift for Kohl's, increased 14% (Peterson 2019). Michelle Gass, Kohl's chief executive officer, notes, "Our top strategic priority is driving traffic, and this transformational program does just that. It drives customers into our stores, and we are expecting millions to benefit from this service" (Kohl's 2019). However, those estimates were based on anonymous location data from mobile devices, and hence, they provide at best indirect estimates of actual foot traffic. Furthermore, they do not reveal whether the traffic was converted into Kohl's sales or whether this improvement was driven by a channel shift of Kohl's consumers from online to stores. Moreover, three weeks of data may be too short to observe the true value of the partnership as consumers take time to become aware of the new return service. Therefore, the value of a return partnership to a location partner is unclear, and its estimation requires a systematic empirical study.

Our primary research objective is to identify whether and, if so, by how much, a return partnership generates additional sales—in the online and store channels—to a location partner. If this effect is statistically significant, it is useful to elucidate the underlying mechanisms that drive the economic value. These mechanisms would then serve as a valuable guide for the various stakeholders as they make strategic decisions regarding their return partnerships.

For our analysis, we have compiled a unique longitudinal data set from multiple sources that span three years (2016 to 2019). Specifically, we have collected novel proprietary data sets from Happy Returns and its major location partner, a stationery and craft omnichannel retailer that has an online sales channel and more than 100 off-line stores across the United States. Our data include the detailed information of a total of 214,225 store returns made for 76 weeks, 32 million transactions at the location partner's store and online channels during 152 weeks, and macroeconomic data that characterize each store location. This retailer introduced the Happy Returns service into its off-line stores at different points in time over one year.

This intertemporal variation in the service initiation provides a quasi-experimental setting to study the impact of the return partnership on the location partner's performance.

We analyze this longitudinal data set using a panel difference-in-differences (DID) model with two-way fixed effects in which a treatment is the initiation of the return service at each of the location partner's stores and an outcome is the performance measure of interest (e.g., number of new customers, sales) in the location partner's off-line and online channels. The results of our analysis provide compelling evidence that a return partnership can improve the performance of a location partner significantly. Our analysis reveals that the return partnership increased the number of unique customers, items sold, and net revenue at off-line stores, on average, by 7.0%, 7.8%, and 6.1%, respectively. Not only did the return partnership improve store performance, but it also increased online sales substantially: in particular, in zip code areas close to—within 10 miles of—the location partner's stores, the number of unique customers, items sold, and net revenue in the online channel increased, on average, by 4.1%, 8.4%, and 7.6%, respectively.

We identify empirical evidence of the two aforementioned mechanisms driving the improvements in sales performance. For the store channel, both customer acquisition and supercharging effects are significant. As expected, the launch of the return service brings in new customers to the location partner's stores—on average, by 4.3% more. Interestingly, the customer supercharging effect is also pronounced: existing customers, on average, made purchases from stores more often (6.7%), purchased more items (6.2%), and generated higher net revenue (5.4%) after their first use of the return service. For the online channel, a major driving force is the customer acquisition effect. Our analysis shows that the return partnership helped the location partner acquire 9.8% more new online customers in zip code areas close to the location partner's stores. We offer two plausible explanations for this effect with empirical support: First, this may be due to the enhanced brand awareness of the location partner through the return service offered to more than 30 other online retailers. Our analysis of Google Trends data shows that the introduction of the return service increased Google searches of the location partner's brand substantially. Second, the convenience of returns may attract more new online customers in the areas where the off-line return option is available by reducing the perceived risk associated with online purchase. Indeed, we find that the value of returns made by new online customers increased significantly more in those areas having the off-line return option. Finally, we find that the customer supercharging effect is not statistically significant in the online channel, suggesting that the online purchase patterns of incumbent customers did not change significantly as a result of using the off-line return service. This ensures that there is no significant channel shift of those customers from online to off-line stores because of the off-line return service.

Our study is timely and important to online and off-line retailers as well as intermediary firms that connect online to off-line retailers. Traditionally, online and off-line retailers view each other as competitors. Recently, we have started to observe the collaborations between online and off-line retailers in processing returns. Our findings can validate the business value of those collaborations and be used to foster more extensive win–win partnerships. We are pleased to report that the results of our study have already been used to help our research partner companies fortify their partnership and to offer strategic guidance into other future opportunities.

2. Literature Review

There is extensive literature that studies return policies for single-channel retailers. Several papers study optimal return policies for such retailers, including Su (2009), Ulku et al. (2013), and Altug and Aydinliyim (2016). There are also a number of empirical papers that study how return policies affect consumers' decisions to return (e.g., Wood 2001, Bechwati and Siegal 2005, Shulman et al. 2015, Shang et al. 2019).

Although this stream of research focuses on a single channel, there is a growing interest in studying omnichannel operations among both analytical and empirical researchers. From a theoretical perspective, Ofek et al. (2011) study competing retailers that can operate both online and off-line channels and examine how pricing strategies and physical store assistance levels change as a result of the additional online channel. Gao and Su (2017b) study how omnichannel retailers can effectively deliver online and off-line information to consumers who strategically choose whether to gather information online or off-line and whether to buy products online or off-line. Gao and Su (2017a) study the buy online, pickup in store (BOPS) option of an omnichannel retailer analytically, and they caution that products that sell well in stores are not well suited for in-store pickup.

Some empirical studies consider the effect of opening retail stores in an omnichannel context. By analyzing a clothing retailer's sales data, Anderson et al. (2010) examine how opening the first retail store in a state affects online and catalog demand because of sales taxes. Wang and Goldfarb (2017) examine how new off-line stores of an omnichannel retailer affect online sales depending on the retailer's brand presence. In their setting, online sales decrease as a result of a new off-line store in locations where the retailer has a strong brand presence. Kumar et al. (2019) show that the opening of off-line stores increases online sales of *existing* customers, and they attribute the improvement to customers' better evaluation of the retailer's product offerings at stores. Our context of a return partnership is different in that it involves a new return service among multiple collaborative firms rather than a new store opening of a single firm, and existing customers of a location partner already have access to stores. As such, different from Kumar et al. (2019), we find that the return service does not induce existing customers to alter their purchase patterns in the online channel of the location partner although it does so in the off-line channel. We also characterize how the return service helps the location partner acquire *new* customers in both off-line and online channels.

Other empirical studies consider the effect of various omnichannel strategies on online and off-line sales. Gallino and Moreno (2014) study the BOPS option of an omnichannel retailer that sells housewares, furniture, and home accessories and find that this option increased traffic and sales of off-line stores while reducing online sales because of an increase of "research online, purchase off-line" behavior. Bell et al. (2018) analyze quasi-experimental data on showroom openings by online-first eyewear retailer WarbyParker.com. They find that showrooms not only increase demand overall and in the online channel, but they provide an influx of new customers in the showroom channel who are predominantly new to the brand. By analyzing individual order–level data of an apparel omnichannel retailer, Bell et al. (2020) find that, after customers visit zero-inventory stores or showrooms, they tend to spend more, shop at a higher velocity, and are less likely to return items, a phenomenon they refer to as "supercharging." The return partnership in our study illustrates a similar supercharging effect for existing customers of the location partner; however, the driving mechanism here is the off-line return service for a different (online) retailer.

There are a few recent studies that focus on product returns of an omnichannel firm. Chen and Chen (2017) examine how an omnichannel retailer should choose between full refunds and no returns. Ertekin and Agrawal (2021) study both analytically and empirically how a change in the return period policy affects sales, returns, and profitability for an omnichannel jewelry retailer. They find that a decrease in the return period window had no statistically significant effect on online sales but reduced store sales. In contrast, our focus is on the impact of providing a new return option for customers of a different (online) retailer. Nageswaran et al. (2020) study the pricing and return policy decisions of an omnichannel retailer serving heterogeneous customers. They attribute the firm's profit improvement from returns to stores partly to cross-selling—customers making additional purchases when they go to off-line stores for returns—in an omnichannel setting, the existence and magnitude of which we empirically quantify in the online–off-line return partnership setting.

The extant research in omnichannel operations focuses on elucidating the interplay between online and off-line stores of a *single* company in purchases and returns. Our paper contributes to this literature by examining the *partnership* between online retailers and location retail partners (intermediated by a third-party company, Happy Returns). We examine how the introduction of an option to drop off returns of items purchased from different online retailers at a location partner's off-line stores influences off-line and online sales for the location partner. We explain the underlying mechanism by characterizing the impact of the return partnership on shopping patterns of existing and new customers in both off-line and online channels. Our result cannot be fully explained by the prior results obtained for a single omnichannel retailer. For example, Gallino and Moreno (2014) show that the BOPS option reduces online sales of an omnichannel retailer because of the significant shift of online customers to stores, but our result shows that the foot traffic induced by the return service does not entail the significant channel shift. A related paper is that by Nageswaran et al. (2021), who analytically study the incentives of an online retailer and a location partner toward a return partnership. Our work complements this study by quantifying a location partner's incentive.

Finally, our paper is also related to a stream of papers that study store traffic and its conversion to sales. There are several studies that examine the impact of marketing activities and price promotions on store traffic (e.g., Lam et al. 2001 and references therein). More recently, Perdikaki et al. (2012) study the relationship between store traffic, labor, and sales performance. They identify the moderating role of labor in the relationship between traffic and sales. Lee et al. (2020) examine the impact of fitting-room traffic on store performance. They find that high traffic to fitting rooms exacerbates stockouts because of unwanted items left in those rooms and demonstrate that dedicated labor can significantly reduce such stockouts. Different from this stream of research, we study collaborative partnerships between omnichannel stores and online retailers and examine how such a partnership affects transaction volume and drives sales via the customer acquisition and supercharging effects.

3. Research Setting and Data

For this study, we collaborated with Happy Returns and its major location partner of which the stores have launched the return service at different points in time from February 2018 to February 2019. At the time of our empirical study, more than 30 online retailers offered the Happy Returns service to their customers. Most of Happy Returns' clients are apparel, footwear, and handbag online retailers, such as Rothy's, Everlane, and Revolve, which are good representatives of online retailers struggling with high return rates. Customers who wish to return online purchases from any of these (partnering) online retailers may visit a location partner's store to do so. Upon arrival to the store, the customer provides the email address associated with the online purchase, a QR code associated with the online-initiated return, or the order number of the original purchase to look up the online order and subsequently drops off the item(s). There is no transaction fee between the location partner and Happy Returns in exchange for offering this service.

We have consolidated a unique longitudinal data set that spans 152 weeks (September 2016 to July 2019) by combining (1) information of off-line returns made at each store of the location partner, (2) transactions at all stores and the online channel of the location partner, and (3) various macroeconomic characteristics of store locations, which we explain in detail next.

The first component of our data set, obtained from Happy Returns, is proprietary data containing information on all the returns made at the stores of the location partner. The return data set covers a 76-week period, starting from the first launch of the Happy Returns service in February 2018 through July 2019. During this period, a total of 214,225 returns were made at the stores of the major location partner with an increasing rate over time as shown in Figure 1. For each return, the data set includes its return date, price of a returned item, customer ID, and the specific location partner's store used for the return. Returned items are mostly fashion items, such as shoes, clothing, bags, watches, and bracelets. The average and median values of returned items are \$107.89 and \$89.00, respectively.

The second component of our data set, obtained from the location partner, is proprietary transaction data (≈32 million transactions) in both off-line and online channels. The transaction data set covers 152 weeks, starting from September 2016 to July 2019, and covers 76 weeks before and after the first launch of the return service in February 2018. For each transaction, the data set includes its purchase date; the channel of the transaction (i.e., off-line or online), including the specific store location for off-line transactions; sales value; customer ID; and zip code. The majority of sales are made in physical stores. During our empirical period, 78.25% of total net revenue is generated from the store channel and the remainder from the online channel.

The last component of our data set contains the macroeconomic characteristics of each store location. There are various characteristics of a store location that may influence store performance. To incorporate them into our estimation model, we collected several zip code–level characteristics of each store location, such as per capita income, population density, education level, gender ratio, and unemployment ratio from the U.S. Census Bureau. We also purchased data of consumer spending on stationery and craft items from Epsilon, a data-driven marketing company, to control for the specific market conditions of the location partner's industry.

4. Econometric Models

We use a DID approach to estimate the economic value of the return partnership; this approach is widely used in operations management (e.g., Bell et al. 2018, 2020; Wang et al. 2021). Leveraging a quasi-experimental setting, a DID model estimates the effect of a treatment on an outcome by comparing changes in the outcome between a population that is affected by the treatment (treated group) and a population that is not (control group) over the same time frame. In our study, the treatment is the launch of the return service at each store of the location partner. We are interested in quantifying the effect of the treatment on store and online channel performance of the location partner.

To estimate the economic value of the return partnership, we rely on the *staggered* launch of the return service. According to Happy Returns, the service rollout schedule was determined based on geographic locations of stores because Happy Returns' team needed to provide training and support to store staff. The first rollout was made as a pilot to the stores in West Los Angeles where Happy Returns' headquarters is. Then, the second rollout was made in stores located in Chicago, the third one in Boston, the fourth in New York, and so on. This region-based rollout was implemented on 10 different dates over seven different weeks between February 2018 and February 2019.

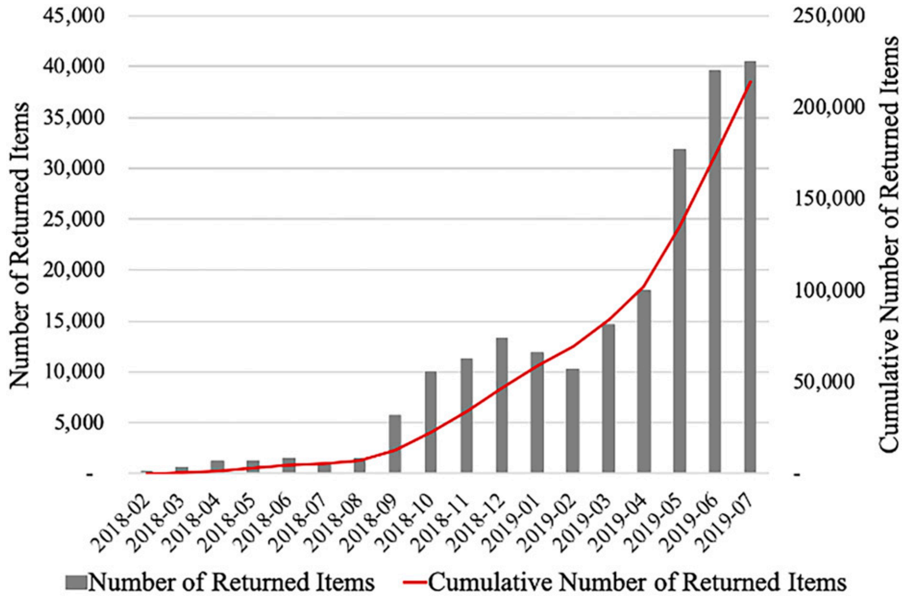
4.1. Model: Store Channel

We estimate the impact of the return partnership on store performance by using the following panel DID model with two-way fixed effects:

$$\begin{aligned}\boldsymbol{Y}_{it} = \beta\, Partnership_{it} + \theta \boldsymbol{X}_{it} + \sum_{i} Store\ fixed\ effects_{i} \\ + \sum_{t} Time\ fixed\ effects_{t} + \sum_{it} Store_{i} * Time_{t} + \varepsilon_{it},\end{aligned} \tag{1}$$

where i is a store index and t is a time (week) index. We use a week for the time unit of our analysis similar to related literature (e.g., Gallino and Moreno 2014,

Figure 1. (Color online) Total Number of Returned Items to Off-line Stores



Bell et al. 2018, Ertekin and Agrawal 2021). Our results are robust to using a month for the time unit of our analysis. Table 1 presents the definition of model variables, and Table 2 provides the descriptive statistics and correlation of model variables. We next explain each term in (1) in detail.

The term $\boldsymbol{Y}_{it}$ in (1) is a vector of three measures of store i's performance during week t: $CUST_{it}$, $QUANT_{it}$, and REV_{it}. The variable $CUST_{it}$ is the total number of unique customers who made purchases at store i during week t. The variable $QUANT_{it}$ is the total number of items sold at store i during week t, and REV_{it} is the net revenue (total revenue minus return value in U.S. dollars) of store i during week t. Throughout the paper, all outcome measures are log-transformed for regressions to correct for their right-skewness. $Partnership_{it}$ is our binary independent variable, which indicates the treatment status about whether store i offered the return service at week t. The coefficient of our main interest, β, is interpreted as the average performance change of store i at week t that is attributable to the return partnership.

Next, to control for potential heterogeneities across stores, we incorporate several covariates into our empirical model. First, we incorporate various time-varying characteristics of the store environment. The vector term $\boldsymbol{X}_{it}$ in (1) contains seven characteristics of the population on the year of week t where store i is located: $PerCapitaIncome_{it}$, $PopulationDensity_{it}$, $UnemploymentRate_{it}$, $PerHouseholdCraftSpending_{it}$, $FamilyHousehold_{it}$, $FemalePopulation_{it}$, and $PopulationBachelorDegree_{it}$. We have chosen these specific characteristics because they are not highly correlated with each other.

Besides controlling for observable characteristics, we also incorporate fixed effects for each store and week to account for unobservable heterogeneities of different stores and weeks, respectively. The store fixed effects, captured by $Store\ fixed\ effects_i$ in (1), tease out the mean differences of performance across different stores that may be driven by their inherent heterogeneities: for example, store size and location (stand-alone shop or within a mall). The time fixed effects, captured by $Time\ fixed\ effects_t$ in (1), control for shocks related to time that are common across stores: for example, seasonality, holiday promotions, and national macroeconomic conditions. We further incorporate store-specific time trends, $\sum_{it} Store_i * Time_t$, to flexibly capture time-varying individual store characteristics that are unobservable to us. For example, the number and quality of store staff, its operating hours, and floor plan may change within a store over time, and those factors may influence store performance. The store-specific time trends can flexibly account for such time-varying factors by allowing unobservable store-specific heterogeneities to change over time. Also, the store-specific time trends can capture macroeconomic conditions that may fluctuate within a year, which cannot be captured by $\boldsymbol{X}_{it}$ as the macroeconomic data are collected on an annual basis. An alternative approach to capturing store-specific heterogeneities is to include a fixed effect for each specific store and time interval. We check the robustness of our results under this alternative specification in the online appendix (Online Tables A11 and A12).

The last term ε_{it} in (1) is an error term. Recent studies demonstrate that regression models produce

Table 1. Definition of Model Variables for the Store Performance Analysis

Variable name	Type	Definition
$CUST_{it}$	DV	Number of unique customers who made purchases in store i during week t
$QUANT_{it}$	DV	Number of items sold at store i during week t
REV_{it}	DV	Net revenue (total revenue − return value) of store i during week t (in U.S. dollars)
$Partnership_{it}$	IV	An indicator variable that takes value of one if store i offered the return service during week t or zero otherwise
$PerCapitaIncome_{it}$	CV	Per capita income (in 10,000 U.S. dollars) in a zip code area where store i is located in the year to which week t belongs
$PopulationDensity_{it}$	CV	Population (in 10,000) per square mile in a zip code area where store i is located in the year to which week t belongs
$UmemploymentRate_{it}$	CV	Unemployment rate in a zip code area where store i is located in the year to which week t belongs
$PerHouseholdCraftSpending_{it}$	CV	Consumer spending amount (in 100 U.S. dollars) on stationery and craft items divided by the number of households in a zip code area where store i is located in the year to which week t belongs
$FamilyHousehold_{it}$	CV	Proportion of family households in a zip code area where store i is located in the year to which week t belongs. Family households are households with two or more members who live in the same home and are related by birth, marriage, or adoption.
$FemalePopulation_{it}$	CV	Proportion of female population in a zip code area where store i is located in the year to which week t belongs
$PopulationBachelorDegree_{it}$	CV	Proportion of population with bachelor's degree in a zip code area where store i is located in the year to which week t belongs.

Note. Variable type: DV stands for a dependent variable, IV for an independent variable, and CV for a control variable.

downward-based standard errors if serial correlation within clusters are not accounted for (Bertrand et al. 2004, Donald and Lang 2007). In our empirical setting, errors are likely to be correlated among observations within the same store or time period. To account for such within-cluster error correlations, we use two-way clustered standard errors at the store and week levels (Angrist and Pischke 2009, Wooldridge 2010).

In a DID model with staggered treatments such as ours, a DID estimator (β in our model) is calculated as a weighted average of all possible two-group/two-period DID estimators in the data (e.g., Goodman-Bacon 2018). Specifically, our estimator is based on comparisons of stores that launched the return service across seven different phases. For each of these phases, stores that started the return service (i.e., changed their treatment status) during this period belong to a treated group, and stores that did not change their treatment status during this period (i.e., had already started the return service or had not yet started

Table 2. Descriptive Statistics of Model Variables for the Store Performance Analysis

	Mean	Standard deviation	Minimum	Maximum	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(1) $CUST_{it}$	415.80	262.58	7.00	2,545.00	1.00										
(2) $QUANT_{it}$	1,789.20	1,172.14	9.80	11,834.00	0.95	1.00									
(3) REV_{it}	16,800.00	10,957.21	5.53	115,946.82	0.93	0.98	1.00								
(4) $Partnership_{it}$	0.34	0.47	0.00	1.00	0.14	0.14	0.11	1.00							
(5) $PerCapitaIncome_{it}$	4.48	2.37	1.81	12.58	0.13	0.16	0.18	0.07	1.00						
(6) $PopulationDensity_{it}$	1.13	2.10	0.06	12.74	0.36	0.28	0.24	0.01	0.16	1.00					
(7) $UmemploymentRate_{it}$	0.04	0.02	0.02	0.11	−0.06	−0.06	−0.08	−0.08	−0.32	−0.06	1.00				
(8) $PerHouseholdCraftSpending_{it}$	2.19	0.73	0.65	11.45	0.25	0.25	0.24	0.00	0.39	0.29	−0.01	1.00			
(9) $FamilyHousehold_{it}$	0.41	0.16	0.21	0.89	−0.06	−0.06	−0.06	0.00	−0.03	0.06	−0.06	−0.27	1.00		
(10) $FemalePopulation_{it}$	0.51	0.02	0.45	0.57	0.06	0.08	0.09	0.01	0.23	0.09	−0.13	0.19	−0.08	1.00	
(11) $PopulationBachelorDegree_{it}$	0.39	0.07	0.12	0.53	0.28	0.27	0.27	0.02	0.44	0.22	−0.36	0.28	−0.15	0.04	1.00

Notes. The unit of analysis is store (i) – week (t). Columns (1)–(11) provide a bivariate correlation between variables. The values of the first three variables are scaled because of the nondisclosure condition.

the return service) belong to a control group. For clarity, we provide the breakdown of treated and control stores in each of these seven phases in Table 3. Table 3 shows that the first treatment happened in the week of February 26, 2018. During this week, eight stores in Los Angeles started the return service. The remaining 118 stores had not started the return service in this phase, so they serve as a control group. Similarly, the second treatment happened in the week of March 12, 2018. Ten stores in Chicago started the return service, so they formed a treated group. As in the first phase, the 108 not-yet-treated stores (i.e., stores that had not started the return service) serve as a control group. In addition, the eight Los Angeles stores that had already started the return service belong to a control group in the second phase. Once estimators are obtained from all comparisons between treated and control stores across seven treatment phases, the final average treatment effect is calculated as a weighted average of the estimators. Similar to any other least squares estimators, the weights are proportional to the sizes of each comparison group and to the share of time it spends treated (see Goodman-Bacon 2018 for details). Note from Table 3 that the stores that stay untreated become zero in the seventh phase as all stores eventually launched the return service during our empirical window. However, our model still has a control group in the seventh phase: the stores that had already started the return service and remained treated during this phase (i.e., 125 stores in the seventh phase). In prior literature that employs a DID model with staggered treatments, it is not uncommon to estimate the average treatment effect without a comparison against never-treated units (e.g., Greenwood and Wattal 2017, Burtch et al. 2018, Babar and Burtch 2020).

4.2. Model: Online Channel

The return partnership can influence online shopping behaviors of customers who reside close to the location partner's stores. Because those customers do not need to travel far to use the return service, they are more likely to use this service and possibly change their shopping behavior—shifting some of their purchasing from online to off-line (i.e., channel-shift effect) or purchasing more online (i.e., supercharging effect)—as a result.

We estimate the impact of the return partnership on online channel performance by using the two-way fixed effects DID model similar to Equation (1), which now uses zip code (z) – week (t) as the level of analysis, ($CUST\_online_{zt}$, $QUANT\_online_{zt}$, REV_online_{zt}) as the dependent variables, and $Partnership_{zt}$ as the independent variable. (See the definition and descriptive statistics of model variables for the online channel in Online Tables A1 and A2.) We calculate the distance between stores and the center of every zip code area where customers reside by using Geocoding API of Google. Our calculation shows that 63.0% of those customers who used the return service reside in zip codes within a five-mile radius of the return location, and 82.7% reside in zip codes within a 10-mile radius. Thus, in our analysis, we define the area of influence of the return service as 10 miles (while also reporting our results with five miles in the online appendix). This means that zip codes within the 10-mile radius from any store that offered the return service at time t are considered treated zip codes, and those outside of the 10-mile radius at time t belong to a control group. This is consistent with what the management team at Happy Returns considers as the area of influence of the return service. Similar to the store performance analysis, the DID estimator of the online channel analysis is calculated based on comparisons of zip codes in treated and control groups over seven different phases (see Table 4). Note, however, that some zip codes are never treated during our empirical period because they are located farther than 10 miles from any of the treated stores. Hence, the DID estimator of the online channel performance also includes a comparison between treated and never-treated zip codes.

We next present the results for the impact of the return partnership on store and online channel performance in Section 5 and conduct several robustness checks in Section 6. Then, Section 7 elaborates on the customer behavior that drives the results. Section 8 concludes our paper.

Table 3. Store Performance Analysis: Breakdown of Treated and Control Stores

Treatment timings			Treated group	Control group	
Phase	Treatment dates	Week starting	Stores that are newly treated	Stores that stay untreated	Stores that stay treated
First	2/26/18, 2/27/18, 2/29/2018	2/26/18	8	118	0
Second	3/13/18, 3/14/18	3/12/18	10	108	8
Third	8/15/18	8/13/18	13	95	18
Fourth	8/29/18	8/27/18	37	58	31
Fifth	9/12/18	9/10/18	32	26	68
Sixth	9/26/18	9/24/18	25	1	100
Seventh	2/11/19	2/11/19	1	0	125

Table 4. Online Performance Analysis: Breakdown of Treated and Control Zip Codes

Treatment timings			Treated group	Control group	
Phase	Treatment dates	Week starting	Zip codes that are newly treated	Zip codes that stay untreated	Zip codes that stay treated
First	2/26/18, 2/27/18, 2/29/2018	2/26/18	221	26977	0
Second	3/13/18, 3/14/18	3/12/18	183	26794	221
Third	8/15/18	8/13/18	311	26483	404
Fourth	8/29/18	8/27/18	1001	25482	715
Fifth	9/12/18	9/10/18	822	24660	1716
Sixth	9/26/18	9/24/18	687	23674	2538
Seventh	2/11/19	2/11/19	299	23973	3225

5. Results: Value of Return Partnership

5.1. Impact of Return Partnership on Store Channel Performance

We present the estimation results of Equation (1) in Table 5. All six models considered in Table 5 include store and time fixed effects, and Models 2, 4, and 6 additionally incorporate store-specific time trends. Because store-specific time trends enhance identification, we use models with these trends to interpret the results throughout this paper. Across all three outcome measures, the coefficient of $Partnership_{it}$ is positive and statistically significant, suggesting a positive impact of the return service on these performance measures of stores. Specifically, the coefficients of $Partnership_{it}$ indicate that the return service helped the location partner attract, on average, 7.0% more customers, sell 7.8% more items, and generate 6.1% higher net revenue. A simple back-of-the-envelope calculation based upon the mean value of store performance given in Table 2 suggests that these numbers are translate to 29 (= 7.0% × 415.80) more customers, 140 (= 7.8% × 1,789.20) more items sold, and \$1,025 (= 6.1% × \$16,800.00) more net revenue per week for a store. Then, the total net revenue impact on all 126 stores amounts to \$129,125 per week.

We conduct two supplementary analyses. First, we examine whether those stores that experience a larger volume of returns after the treatment observe a larger increase in sales. For this analysis, we use the same specification as (1) although replacing $Partnership_{it}$ with $ReturnCount_{it}$, which represents the total number of returns that store i received via the partnership at time t. The result of this analysis (presented in Online Table A3) shows that the coefficient of $ReturnCount_{it}$ is positive and statistically significant, confirming the intuition that the sales of those stores that received more returns via the partnership indeed increased more. Second, we examine how the availability of other stores in the neighborhood that offer the same return service affects the benefits of the return partnership. To examine this potential "competition" effect, we construct a variable $OtherReturnLocations_{it}$, which captures the number of other Happy Returns locations that are in the vicinity (i.e., within 10 miles) of store i at week t. In addition, we introduce an interaction term $Partnership_{it} \times OtherReturnLocations_{it}$ to investigate whether the number of other return locations moderates the effect of the return partnership on store performance. The results of this analysis (presented in Online Table A4) show that the coefficient of $OtherReturnLocations_{it}$ is positive and significant, indicating that stores in the area with more neighboring return locations tend to have higher net revenue on average. The coefficient of $Partnership_{it} \times Other ReturnLocations_{it}$ is negative and statistically significant. This indicates that there is indeed the competition effect: the presence of more stores in the vicinity that offer the same return service reduces the benefits of the partnership. However, it turns out that the magnitude of this competition effect is not substantial as indicated by the low values of the estimated coefficients ($\beta = -0.004$, $p < 0.001$ for $DV = log(Rev_{it})$).

5.2. Impact of Return Partnership on Online Channel Performance

In Section 5.1, we find that the store channel performance improved as a result of the return partnership. At this point, one may ask how the online channel performs: does it show substantial improvement as well, or is there a channel shift driving the improvement in the store channel performance? Gallino and Moreno (2014) show that, although the BOPS option of an omnichannel retailer increased store traffic, it reduced online sales because of a significant channel shift as a result of customers' research online, purchase off-line behavior, whereas Kumar et al. (2019) show that a new store opening increases online sales. Our setting—involving interactions in multiple channels and multiple retailers rather than a stand-alone omnichannel firm—is quite different from these studies, making it interesting to investigate the impact of the return partnership on online sales, which is the focus of this section.

Table 6 presents the results of our analysis of the return partnership's impact on the online channel performance. The results indicate that the coefficients of

Table 5. Impact of Return Partnership on Store Channel Performance

Dependent variable	$log(CUST_{it})$		$log(QUANT_{it})$		$log(REV_{it})$	
Model	(1)	(2)	(3)	(4)	(5)	(6)
Treatment						
$Partnership_{it}$	0.082***	0.070***	0.082***	0.078***	0.065***	0.061***
	(0.017)	(0.009)	(0.017)	(0.011)	(0.017)	(0.011)
Controls						
$PerCapitaIncome_{it}$	0.005***	0.006***	0.003***	0.001**	0.005***	0.001**
	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)
$PopulationDensity_{it}$	0.090***	0.087***	0.067***	0.059***	0.054***	0.046***
	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)
$UmemploymentRate_{it}$	−0.377***	−0.312***	−0.303***	−0.231***	−0.334***	−0.244***
	(0.020)	(0.011)	(0.019)	(0.010)	(0.019)	(0.011)
$PerHouseholdCraftSpending_{it}$	0.126***	0.146***	0.130***	0.104***	0.126***	0.100***
	(0.005)	(0.003)	(0.004)	(0.002)	(0.004)	(0.002)
$FamilyHousehold_{it}$	−0.178***	−0.140***	−0.149***	−0.026**	−0.121***	−0.118***
	(0.013)	(0.008)	(0.013)	(0.007)	(0.013)	(0.007)
$FemalePopulation_{it}$	0.078***	0.075***	0.119***	0.109***	0.176***	0.112***
	(0.016)	(0.009)	(0.016)	(0.008)	(0.016)	(0.007)
$PopulationBachelorDegree_{it}$	0.939***	0.938***	0.909***	0.893***	0.911***	0.898***
	(0.024)	(0.017)	(0.024)	(0.016)	(0.024)	(0.017)
Store fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Store-specific time trends	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.832	0.849	0.841	0.871	0.879	0.901

Notes. The unit of analysis is store (i) – week (t). The number of observations is 19,152. Numbers in parentheses are two-way clustered standard errors at store and week level.
$^*p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.001$.

$Partnership_{zt}$ are positive and significant for all outcome measures of online sales. Specifically, the return partnership increased the average number of online customers, online sale quantity, and net revenue by 4.1%, 8.4%, and 7.6%, respectively. Based on the mean value of online performance, these numbers are translated to 0.11 more customers, 0.98 more items, and $15.92 more net revenue per week for each affected zip code. Then, the total net revenue impact on all 3,524 affected zip codes amounts to $56,109 per week.

These results suggest that the return partnership attracts customers and increases sales in the online channel as well. Moreover, this result implies that the enhanced store performance resulting from the return partnership we observed in Section 5.1 is unlikely to be driven by a channel shift effect. This is because we would observe an online sales drop after the return partnership if the enhanced store performance were driven primarily by a channel-shift behavior (i.e., online customers switch their purchase channel to off-line).

6. Robustness Checks

The validity of a DID estimator depends on the parallel trends between treated and control groups (e.g., Bertrand et al. 2004, Angrist and Pischke 2009) because the treatment effect is identified based on the assumption that the posttreatment trend of a control group is a counterfactual of a treated group's trend without treatment. Because treatment is not randomly assigned in a quasi-experimental setting, those units that receive treatment (treated group) may be inherently different from those that do not get treatment (control group). If such a pretreatment difference exists, estimates from a quasi-experiment may be confounded. For example, if high-performing stores were chosen to launch the partnership earlier than others, our estimate might be confounded, and the observed performance increase after the partnership might have been driven by the treated stores' capability to achieve high performance gains and not by the return partnership. The fact that launch timings were determined based on a store's geographic location (i.e., metropolitan area) rather than on an individual store's performance is noteworthy as it suggests that the treatment timing is not assigned endogenously. Although the parallel trends assumption is inherently untestable, we conduct several robustness checks in this section to bolster our confidence in the plausibility of parallel trends.

To begin, we visually check whether treated and control units follow similar performance trends in pretreatment periods. As shown in Tables 3 and 4, stores and zip codes belonging to treated and control groups differ in seven different treatment phases. Thus, for each treatment phase, we plot the pretreatment trends of store and online net revenues over the one-year period (March 2017−February 2018) prior to

Table 6. The Impact of Return Partnership on Online Channel Performance

Dependent variable	$log(CUST\_online_{zt})$		$log(QUANT\_online_{zt})$		$log(REV\_online_{zt})$	
Model	(1)	(2)	(3)	(4)	(5)	(6)
Treatment						
$Partnership_{zt}$	0.044***	0.041***	0.126***	0.084***	0.109***	0.076***
	(0.004)	(0.002)	(0.008)	(0.005)	(0.006)	(0.004)
Controls						
$PerCapitaIncome_{zt}$	0.061***	0.057***	0.059**	0.034***	0.023***	0.019**
	(0.001)	(0.001)	(0.028)	(0.003)	(0.004)	(0.004)
$PopulationDensity_{zt}$	0.091***	0.090***	0.147**	0.143**	0.071***	0.053***
	(0.018)	(0.017)	(0.048)	(0.049)	(0.014)	(0.006)
$UmemploymentRate_{zt}$	−0.605***	−0.407***	−0.616***	−0.473***	−0.600**	−0.342***
	(0.070)	(0.071)	(0.188)	(0.019)	(0.248)	(0.025)
$PerHouseholdCraftSpending_{zt}$	0.075***	0.060***	0.083***	0.069***	0.097***	0.072***
	(0.001)	(0.007)	(0.002)	(0.002)	(0.002)	(0.001)
$FamilyHousehold_{zt}$	−0.078***	−0.076***	−0.072**	−0.066***	−0.050***	−0.047***
	(0.012)	(0.011)	(0.032)	(0.028)	(0.004)	(0.004)
$FemalePopulation_{zt}$	0.198***	0.175**	0.519***	0.455**	0.398**	0.314**
	(0.070)	(0.081)	(0.161)	(0.218)	(0.186)	(0.154)
$PopulationBachelorDegree_{zt}$	0.678***	0.503***	0.509***	0.469***	0.504**	0.398***
	(0.059)	(0.052)	(0.137)	(0.139)	(0.183)	(0.118)
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Zip code–specific time trends	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.861	0.865	0.867	0.873	0.893	0.914

Notes. The unit of analysis is zip code (z) – week (t). The number of observations is 409,900. Numbers in parentheses are two-way clustered standard errors at zip code and week level.
$^*p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.001$.

the first treatment in late February 2018, respectively. From Online Figure A1, we can check visually that the difference of net revenue between treated and control groups shows parallel trends over time, which offers suggestive evidence supporting the parallel trends. In the following, we further investigate parallel trends by conducting a lead-and-lags analysis in Section 6.1, a random implementation test in Section 6.2, and a coarsened exact matching analysis in Section 6.3.

6.1. Leads-and-Lags Analysis

In this section, we conduct a leads-and-lags analysis. The leads-and-lags analysis, proposed by Autor (2003), enables us to examine intertemporal dynamics of the treatment effect by replacing a treatment indicator variable with multiple indicator variables, each representing a different period relative to a treatment timing. Dummy variables indicating periods prior to a treatment (pretreatment placebos) allow us to check the parallel trends assumption by quantifying whether there is a statistically significant difference in their outcomes between treated and control groups before a treated group gets the treatment. In addition, through the dummy variables indicating time periods after a treatment, we can examine the temporal dynamics of the treatment effect. One week prior to the return partnership is set as a baseline. Hence, the leads-and-lags estimates represent relative effects of the return partnership to the baseline.

For the store channel, we conduct a leads-and-lags analysis using the following equation:

$$\begin{aligned}\boldsymbol{Y}_{it} = {}& \beta_1 PrePartnership\,[\le -25]_{it} \\ &+ \beta_2 PrePartnership\,[-24, -21]_{it} \\ &+ \beta_3 PrePartnership\,[-20, -17]_{it} \\ &+ \beta_4 PrePartnership\,[-16, -13]_{it} \\ &+ \beta_5 PrePartnership\,[-12, -9]_{it} \\ &+ \beta_6 PrePartnership\,[-8, -5]_{it} \\ &+ \beta_7 PrePartnership\,[-4, -2]_{it} \\ &+ \beta_8 PostPartnership\,[0, 4]_{it} \\ &+ \beta_9 PostPartnership\,[5, 8]_{it} \\ &+ \beta_{10} PostPartnership\,[9, 12]_{it} \\ &+ \beta_{11} PostPartnership\,[13, 16]_{it} \\ &+ \beta_{12} PostPartnership\,[17, 20]_{it} \\ &+ \beta_{13} PostPartnership\,[21, 24]_{it} \\ &+ \beta_{14} PostPartnership\,[\ge 25]_{it} \\ &+ \theta \boldsymbol{X}_{it} + \sum_i Store\ fixed\ effects_i \\ &+ \sum_t Time\ fixed\ effects_t + \sum_{it} Store_i * Time_t + \varepsilon_{it}.\end{aligned} \tag{2}$$

In (2), the dependent variable $\boldsymbol{Y}_{it}$ captures store i's performance at week t. The indicator variable $PrePartnership[\le -25]_{it}$ is coded as one if time t

precedes the service launch at store i by 25 weeks or longer. We then create dummy variables in four-week intervals: $PrePartnership[-24, -21]_{it}$ is coded as one if time t precedes the service launch at store i by 21 weeks to 24 weeks, $PrePartnership[-20, -17]_{it}$ is coded as one if time t precedes the service launch at store i by 17 weeks to 20 weeks, and so on. Additionally, the posttreatment period is coded as follows: $PostPartnership[0,4]_{it}$ captures the effect of the return service during the first four weeks, $PostPartnership[5,8]_{it}$ captures the effect from the fifth to the eighth week, and so on. $PostPartnership[\geq 25]_{it}$ captures the effect from 25 weeks on.

We report the results of the leads-and-lags analysis for the store channel in Figure 2 and Table 7. The coefficients of pretreatment variables are statistically insignificant, indicating that, in the absence of the return partnership (before a treated group gets the treatment), the changes in store performance are not statistically different between treated and control stores. Although this is not definitive evidence of parallel trends, it provides reassuring patterns. These results are consistent when we use time intervals different from four weeks. The coefficients of posttreatment variables indicate that the positive effect of the return partnership on store performance is lagged by nine weeks: the coefficient of $PostPartnership[0,4]_{it}$ is not statistically significant and that of $PostPartnership[5,8]_{it}$ is statistically significant only at a p-value of the 0.1 level. The coefficients stay significant (at a p-value of 0.05) from the ninth week after the service launch, suggesting that the effect of the return partnership persists. These results indicate that the positive effect of the return partnership does not materialize immediately, but it remains as customers become aware of the convenient off-line return option.

Figure 2. (Color online) Impact of Return Partnership on Store Revenue over Time

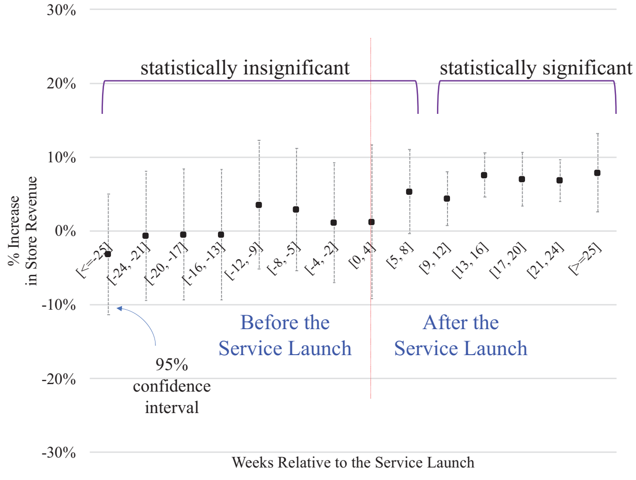


Table 7. Impact of Return Partnership on Store Performance: Leads-and-Lags Analysis

	$log(CUST_{it})$	$log(QUANT_{it})$	$log(REV_{it})$
$PrePartnership\ [<=-25]_{it}$	−0.067	−0.062	−0.032
	(0.042)	(0.041)	(0.042)
$PrePartnership\ [-24, -21]_{it}$	−0.031	−0.031	−0.006
	(0.045)	(0.045)	(0.045)
$PrePartnership\ [-20, -17]_{it}$	−0.018	−0.024	−0.005
	(0.045)	(0.045)	(0.045)
$PrePartnership\ [-16, -13]_{it}$	−0.018	−0.032	−0.005
	(0.045)	(0.045)	(0.045)
$PrePartnership\ [-12, -9]_{it}$	0.008	0.009	0.035
	(0.044)	(0.044)	(0.044)
$PrePartnership\ [-8, -5]_{it}$	0.020	0.023	0.029
	(0.042)	(0.042)	(0.042)
$PrePartnership\ [-4, -2]_{it}$	0.013	0.018	0.011
	(0.042)	(0.041)	(0.042)
$PostPartnership\ [0, 4]_{it}$	0.017	0.005	0.012
	(0.040)	(0.040)	(0.053)
$PostPartnership\ [5, 8]_{it}$	0.082*	0.073*	0.053*
	(0.044)	(0.037)	(0.029)
$PostPartnership\ [9, 12]_{it}$	0.078**	0.069***	0.043**
	(0.032)	(0.015)	(0.019)
$PostPartnership\ [13, 16]_{it}$	0.096***	0.085***	0.076***
	(0.025)	(0.021)	(0.015)
$PostPartnership\ [17, 20]_{it}$	0.113***	0.111***	0.070***
	(0.030)	(0.015)	(0.018)
$PostPartnership\ [21, 24]_{it}$	0.091***	0.108***	0.068***
	(0.014)	(0.024)	(0.014)
$PostPartnership\ [\geq 25]_{it}$	0.105***	0.117***	0.079***
	(0.022)	(0.029)	(0.027)
Control variables	Yes	Yes	Yes
Store fixed effects	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes
Store-specific time trends	Yes	Yes	Yes
Adjusted R^2	0.849	0.871	0.901

Notes. The unit of analysis is store (i) – week (t). The number of observations is 19,152. Numbers in parentheses are two-way clustered standard errors at store and week level.
$^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.001$.

Our result also cautions the interpretation of the result in Peterson (2019) (mentioned in Section 1), which compares foot traffic to Kohl's stores in the three weeks before and after the launch of the Amazon returns program: their effect could have been different if they had extended the analysis longer than three weeks.

We also conduct a leads-and-lags analysis for the online channel, using the same equation as (2) except that the level of analysis changes to zip code (z) – week (t). The results of this analysis, presented in Online Table A5, show no evidence that the parallel trends assumption is violated.

6.2. Random Implementation Test

In this section, we conduct a random implementation test to further check the validity of the parallel trend assumption. The primary objective of this test is to examine how likely we are to observe a significant effect from a random implementation of the treatment. If the observed treatment effect is not driven by the treatment (i.e., the return partnership), but rather by the systematic difference between treated and control groups, we would still observe the treatment effect even with any random draws for the treatment dates. In contrast, if the observed effect is driven by the treatment and not by the idiosyncrasies of two groups, we would observe an insignificant treatment effect when using the random treatment dates.

We conduct this random implementation test as follows. First, we randomly assign treatment dates to stores. After randomly shuffling treatment timings across stores, we replicate our main analysis using the DID model in Equation (1). We conduct this replication 1,000 times and store the estimated coefficients in each iteration. We present the distribution of the 1,000 estimated coefficients from the random implementation test in Figure 3. We observe that the mean of those pseudo-treatment coefficients is statistically indistinguishable from zero in both store and online channel analyses. We further calculate the *Z*-score that measures the statistical difference between our estimated coefficient based on the actual treatment dates ("actual estimate") and the mean of the estimated coefficients based on the pseudo treatments ("pseudo estimate"). The *Z*-scores are 5.48 ($p < 0.001$) and 6.74 ($p < 0.001$) for the store and online channel, respectively, indicating that the actual estimates are significantly larger than the respective pseudo estimates.

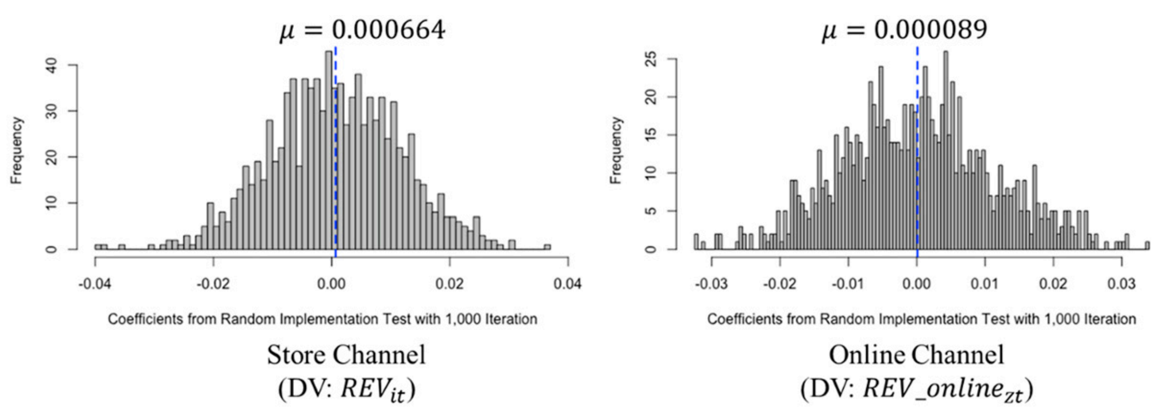
In sum, the results of this random implementation test suggest that our estimates of store and online performance are unlikely to be driven by systematic differences between treated and control groups. We have also conducted this random implementation test for the other two outcome variables—the number of items sold (*QUANT*) and the number of unique customers (*CUST*)—and found the pseudo treatment effects are insignificant as well.

6.3. Coarsened Exact Matching

To check the robustness of our results, we refine the comparability of treated and control groups by using the coarsened exact matching approach (Iacus et al. 2012). In the case of a DID model with staggered treatments, treated and control groups change across different treatment phases (as shown in Tables 3 and 4). Hence, we conduct matching for *each* treatment phase: for each phase, treated units are kept intact, and control units are pruned if they do not have a comparable treated unit. By pruning those control units that are not good counterfactuals to their corresponding treated groups, we can enhance the comparability of treated and control groups in each treatment phase. In this process, the nonparametric coarsened exact matching technique offers a more flexible way to identify comparable treated and control units than the parametric propensity score matching approach (Rosenbaum and Rubin 1983) because it can consider both univariate and multivariate imbalances (e.g., Iacus et al. 2012, Dutt and King 2014).

For our analysis of the store channel, we use the past performance of stores and the various sociodemographic characteristics of store locations to match treated and control groups; see Online Table A6 for the complete list of variables used for this matching. Using the package "cem" of R programming language (Iacus et al. 2018), we coarsen the variables to algorithmically determined buckets and then conduct an exact match to identify comparable treated and control units. The outcome of matching is assessed based on a $\mathcal{L}_1$ statistic (Iacus et al. 2009, 2012). This statistic measures global imbalance, and it varies in $[0, 1]$ where zero indicates perfect global balance and one indicates complete imbalance. The value of $\mathcal{L}_1$ is not useful on

Figure 3. (Color online) Distribution of Coefficients from Random Implementation Test with 1,000 Replications



its own, but as a point of comparison between prematched and postmatched data. An effective matching would reduce the $\mathcal{L}_1$ statistic. For our store channel, the $\mathcal{L}_1$ statistic before the matching is 0.35, and the statistic after the matching is 0.28, indicating that the coarsened exact matching improves comparability between the treated and control groups. We also generate Table 8 that shows the balance between the treated and control groups of stores before and after matching. We find that the mean performance of the treated group and that of the control group in all treatment phases are statistically indifferent after matching with lower values of t-statistics than before matching.

For estimation, we use the same model as Equation (1) except that we do not include $\boldsymbol{X}_{it}$ because the treated and control groups are already matched based on these control variables. (For a robustness check, we also ran a model including $\boldsymbol{X}_{it}$ using the matched data and found that its coefficients are no longer statistically significant.) Because the coarsened exact matching may produce varying numbers of control group stores to each treated store, we weigh each observation based on the size of the stratum to which it belongs and make the number of treated and control groups equal in each stratum (Iacus et al. 2009). The results of this estimation, presented in Online Table A7, show that the results are qualitatively similar to those reported in Table 5 that use the unmatched data.

We conduct a similar analysis for the online channel. We match comparable zip codes by using the sociodemographic characteristics of each zip code. The matching reduces the $\mathcal{L}_1$ statistic from 0.49 to 0.31 and enhances the balance between the treated and control groups in each treatment phase substantially. The estimation results are also consistent with those of Table 6. See Online Tables A8 and A9 for details.

7. Mechanisms That Drive the Value of the Return Partnership

The previous section demonstrates how much the return partnership affects store and online channel performances. In this section, we explore two mechanisms that may drive the enhanced performance: customer acquisition effect in Section 7.1 and customer supercharging effect in Section 7.2.

7.1. Customer Acquisition Effect

The customer acquisition effect may arise when the presence of the new return option brings in new customers, and subsequently, these customers purchase the location partner's products from its off-line and online channels. To investigate this effect, we identify new customers from the acquisition dates of customers (i.e., the date when a customer made the customer's first purchase either in a store or online) the location provider provided to us. We consider a customer a new customer at any week t, if the customer's acquisition date belongs to week t.

First, we investigate how the return partnership helps the location partner acquire new customers in its *store* channel. From descriptive statistics, we find that 106,206 customers returned at least one item purchased from Happy Returns' online retail clients to the location partner's stores during our empirical period. Out of those customers, 16.8% purchased at least one item from stores, and out of those customers, 29.3% were new customers. Although this provides initial evidence, we formally investigate the customer acquisition effect in the store channel. For estimation, we employ the same model as Equation (1) except that the outcome variable is changed to $\boldsymbol{Y\_new}_{it}$, which includes three outcome measures of new customers: $CUST\_new_{it}$, $QUANT\_new_{it}$, and REV_new_{it}. The variable $CUST\_new_{it}$ represents the total number of new customers acquired in store i at week t, $QUANT\_new_{it}$ the total number of items purchased by new customers in store i at week t, and REV_new_{it} the total net revenue generated by new customers in store i at week t. The results of this analysis are presented in Table 9. The coefficients of $Partnership_{it}$ for all outcome variables become significant at p-value 0.05 level in Models 2, 4, and 6 after we incorporate

Table 8. Average Weekly Net Revenue per Store: Balance Between Treated and Control Stores

Treatment timing	Treated stores (Before matching)	Control stores (Before matching)	| *t-stat* |	Treated stores (After matching)	Control stores (After matching)	| *t-stat* |
First	\$20,911.33	\$16,347.69	2.43**	\$20,911.33	\$20,872.68	0.09
Second	\$18,587.65	\$18,427.87	1.47	\$18,587.65	\$18,586.76	0.05
Third	\$15,254.35	\$16,035.02	0.84	\$15,254.35	\$15,231.64	0.04
Fourth	\$16,880.12	\$16,587.28	0.16	\$16,880.12	\$16,587.28	0.16
Fifth	\$16,251.34	\$15,885.10	0.41	\$16,251.34	\$16,283.88	0.10
Sixth	\$14,293.20	\$16,935.29	1.85	\$14,293.20	\$14,286.51	0.08
Seventh	\$20,596.63	\$20,345.75	1.16	\$20,596.63	\$20,574.64	0.05

Notes. The statistics are based on one-year store performance prior to the first treatment date (February 26, 2018). The revenue values are scaled because of the nondisclosure condition.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.001$.

Table 9. Impact of Return Partnership on New Customer Acquisition in the Store Channel

Dependent variable	$log(CUST\_new_{it})$		$log(QUANT\_new_{it})$		$log(REV\_new_{it})$	
Model	(1)	(2)	(3)	(4)	(5)	(6)
$Partnership_{it}$	0.0458*	0.043**	0.072*	0.068**	0.063*	0.061***
	(0.024)	(0.018)	(0.037)	(0.027)	(0.034)	(0.013)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Store fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Store-specific time trends	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.626	0.639	0.679	0.691	0.714	0.756

Notes. The unit of analysis is store (i) – week (t). The number of observations is 19,152. Numbers in parentheses are two-way clustered standard errors at store and week level.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.001$.

store-specific time trends. The results indicate that the launch of the return service increased the number of new store customers, their purchase quantity and sales, on average, by 4.3%, 6.8%, and 6.1%, respectively. This result is consistent with our initial observation from the descriptive statistics.

Next, we examine how the return partnership helps the location partner acquire new customers in its *online* channel. Similar to our analysis in Section 5.2, we estimate this impact at the zip code–week level, at which the zip codes within (outside) the 10-mile radius from any store that offered the return service belong to the treated (control) group. The outcome variable is now $\boldsymbol{Y\_new\_online}_{zt}$, which is a vector of measures of online sales (including the number of new customers, quantity, and net revenue) at week t from new customers who reside in zip code z. Table 10 presents the results of this analysis, showing that the return partnership increased the number of new online customers, their online purchase quantity and sales, on average, by 9.8%, 19.9%, and 20.4%, respectively.

Although the observed increase in new online customers could be due to a variety of different reasons, we offer two plausible explanations. First, the presence of the new return option brings in new online customers who become familiar with the location partner's brand. More than 30 online retailers use the return service, and they publicize how the off-line return service works on their websites, listing various local stores of the location partner. Thus, customers of the partnered online retailers who reside near the location partner's stores are exposed to the location partner's brand, which may translate into a greater influx of new online customers in the affected areas.

To empirically investigate this mechanism, we collected Google Trends data. Google Trends is a service offered by Google that provides how frequently a given search term is entered into Google's search engine in a specific region and time. Because of the vast amount of web searches occurring every day, Google Trends data are frequently used to track brand awareness (e.g., Armstrong 2016) and to predict near-term economic indicators (e.g., Choi and Varian 2012). For our analysis, we collected the weekly search frequency of the location partner's brand name over the same period of our existing data (from September 2016 to July 2019). We collected weekly search data across different geographic regions because we need data on how search frequencies differentially change in the affected versus unaffected areas after introducing the return partnership. Because the most granular region unit that Google Trends data offer is a designated market area (DMA), our data are collected for each

Table 10. Impact of Return Partnership on New Customer Acquisition in the Online Channel

Dependent variable	$log(CUST\_new\_online_{zt})$		$log(QUANT\_new\_online_{zt})$		$log(REV\_new\_online_{zt})$	
Model	(1)	(2)	(3)	(4)	(5)	(6)
$Partnership_{zt}$	0.111***	0.098***	0.213***	0.199***	0.241***	0.204***
	(0.007)	(0.002)	(0.009)	(0.004)	(0.011)	(0.007)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Zip code–specific time trends	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.647	0.654	0.668	0.669	0.705	0.710

Notes. The unit of analysis is zip code (z) – week (t). The number of observations is 409,900. Numbers in parentheses are two-way clustered standard errors at zip code and week level.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.001$.

DMA–week pair. A DMA is a region in which the population receives similar television and radio station offerings and is used in measuring audience measurements by Nielsen Media Research (Nielsen 2021).

To quantify the impact of the return partnership on Google search frequency, we use the same DID model that we used for our online channel analysis except that our unit of analysis is DMA–week level instead of zip code–week level. The outcome variable of this analysis is $GoogleSearch_{dt}$, where d is a DMA index and t is a time (week) index. $Partnership_{dt}$ indicates the treatment status: if at least one store in DMA d offered the return service in week t, $Partnership_{dt}$ takes the value of one and otherwise zero. The results of this analysis are reported in Table 11. The coefficient of $Partnership_{dt}$ indicates that the introduction of the return service increased Google searches of the location partner's brand by 10.4% more in the affected than the unaffected areas. This result suggests that the return partnership increases consumers' awareness of the location partner's brand, which may lead to a greater influx of new online customers and subsequent sales from them.

Second, the convenience of returns may attract more new online customers in the affected zip code areas than the unaffected zip code areas. Returns are easier for customers who live close to partnered stores. This convenience is likely to lower their perceived risk associated with online purchase of the location partner's products, which may be translated into a greater influx of new online customers in the affected areas (i.e., areas within 10 miles of stores offering the Happy Returns service). To explore this mechanism, we first check whether new online customers in the affected areas tend to return products through a store channel and find that they do: out of all new online customers who live within 10 miles of any store that offered the return service, 99.32% made at least one return to a store during our empirical period. Next, we examine whether returns from new online customers increase after the launch of the return service. We employ the same DID model as before while changing the outcome variable to $ReturnValue_{zt}$. The outcome variable $ReturnValue_{zt}$ captures the total dollar value of the items returned by new online customers residing in zip code z at week t. (Note that we do not use the total number of returned items as our outcome variable because we do not have item-level return data.) In this model, we also control for the total sales of new online customers in zip code z at week t because, when the sales amount increases, the total return value tends to increase. The results of this analysis are presented in Table 12. Consistent with our expectation, we find that returns by new online customers increased at a higher rate in the affected areas than in the unaffected areas. In particular, the value of returns made by new online customers in the affected zip code areas increased by 6.7% compared with that in the unaffected areas. This increase is more substantial than the increase in the value of returns made by new store customers ($\beta = 0.0124$, $p < 0.05$); the return partnership has no significant effect on the value of returns made by existing customers. These results suggest that the return convenience offers one plausible explanation for the new customer acquisition in the online channel.

7.2. Customer Supercharging Effect

The customer supercharging effect arises when existing customers of the location partner purchase more and/or more frequently with the location partner as a result of their adoption of the return service. We refer to "existing customers" of the location partner as those customers who purchased at least one item before the first launch of the return service in February 2018. Different from the previous sections, this section presents a customer-level analysis that focuses on the shopping pattern change of existing customers. We conduct this analysis at a customer level because customers started to use the return service at different points in time. A store-level analysis would be

Table 11. Impact of Return Partnership on Google Search Frequencies of the Location Partner's Brand

Dependent variable	$\log(GoogleSearch_{dt})$
$Partnership_{dt}$	0.104***
	(0.024)
Control variables	Yes
DMA fixed effects	Yes
Time fixed effects	Yes
DMA-specific time trends	No
Adjusted R^2	0.638

Notes. The unit of analysis is DMA (d) – week (t). The number of observations is 31,920. Numbers in parentheses are two-way clustered standard errors at DMA and week level.
$^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.001$.

Table 12. Impact of Return Partnership on New Online Customers' Returns

Dependent variable	$\log(ReturnValue_{zt})$
$Partnership_{zt}$	0.067***
	(0.009)
Control variables	Yes
Zip code fixed effects	Yes
Time fixed effects	Yes
Zip code–specific time trends	No
Adjusted R^2	0.631

Notes. The unit of analysis is zip code (z) – week (t). The number of observations is 409,900. Numbers in parentheses are two-way clustered standard errors at zip code and week level.
$^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.001$.

problematic as it assumes all existing customers of a specific store start using the return service on the same day (i.e., the day when the store is treated).

We first investigate whether the return service altered the shopping patterns of the location partner's existing customers in the *store* channel. We hypothesize that, after using the return service, existing customers may purchase more items from the location partner's stores. With the return service being offered to customers, there is an additional reason for them to visit the stores. If they visit stores more frequently for off-line returns, they may buy more items from stores. An alternative hypothesis is that, if existing customers use the return service only during their regular trips to stores for buying other items, they may not necessarily purchase more items from those stores. In this case, the cross-selling benefit to stores may not be significant.

As a preliminary analysis, we first check our data to find out how often existing customers made purchases while visiting the return locations. We find that existing customers who have started to use the return service purchased at least one item during most (77.69%) of their return visits to stores. To formally test our hypothesis, we investigate if purchase frequency, purchase quantity, and spending amount of existing customers changed after their first use of the return service. The staggered adoption of the return service by existing customers enables us to identify the effect of the return service on their purchase behavior. In this analysis, the treated group is a group of existing customers who used the return service at least once, and a control group is a group of existing customers who did not. For this estimation, we use the following DID model:

$$\begin{aligned} \boldsymbol{Y}_{jt} = \beta ServiceUse_{jt} &+ \sum_{j} Customer\ fixed\ effects_j \\ &+ \sum_{t} Time\ fixed\ effects_t + \sum_{jt} Customer_j * Time_t + \varepsilon_{jt}, \end{aligned} \tag{3}$$

where j is an index for an existing customer. The term $\boldsymbol{Y}_{jt}$ includes three outcome measures: customer j's purchase frequency (in days), purchase quantity, and spending amount at week t. The term $ServiceUse_{jt}$ indicates whether customer j used the return service at or before week t. The individual fixed effect helps us tease out the mean differences between the treated and control groups by capturing inherent individual heterogeneity of customer j that is not observable to us. The time fixed effects and customer-specific time trends are defined similarly to our earlier analyses.

Table 13 presents our results. The positive and statistically significant coefficients of $ServiceUse_{jt}$ suggest that, after their first use of the return service, existing customers made purchases more frequently (6.7%), purchased more items (6.2%), and generated higher net revenue (5.4%) for the location partner. This result serves as evidence for the presence of the customer supercharging effect in the store channel resulting from the return service.

We next investigate whether the return service altered the shopping patterns of the location partner's existing customers in the *online* channel. For this analysis, we use the same model as Equation (3) and conduct the analysis at the customer (j) and week (t) levels. We modify outcome variables to $\boldsymbol{Y}\_online_{jt}$, where $\boldsymbol{Y}\_online_{jt}$ includes the following three measures of online purchases of an existing customer j at week t: online purchase frequency (in days), online purchase quantity, and online spending amount at week t.

From this analysis, we can also verify whether the enhanced store performance is driven by reduced online sales to accommodate the possibility of a channel shift for existing customers. It is plausible that those customers who purchased additional items during their return trips to off-line stores could have purchased the same items online if they had not visited stores to use the return service. If such a channel shift of existing customers from online to off-line stores is strong, then we may see negative and significant coefficients of $ServiceUse_{jt}$.

Table 13. The Impact of Return Partnership on Existing Customers' Store Purchase Behavior

Dependent variable	$log(FREQUENCY_{jt})$		$log(QUANT_{jt})$		$log(REV_{jt})$	
Model	(1)	(2)	(3)	(4)	(5)	(6)
$ServiceUse_{jt}$	0.068***	0.067***	0.064**	0.062***	0.056**	0.054***
	(0.010)	(0.006)	(0.025)	(0.010)	(0.026)	(0.009)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Customer fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Customer-specific time trends	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.634	0.638	0.704	0.708	0.736	0.739

Notes. The unit of analysis is customer (j) – week (t). The number of observations is 5,562,207. Numbers in parentheses are two-way clustered standard errors at customer and week level.

$^*p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.001$.

Table 14. The Impact of Return Partnership on Existing Customers' Online Purchase Behavior

Dependent variable	$log(FREQUENCY\_online_{jt})$		$log(QUANT\_online_{jt})$		$log(REV\_online_{jt})$	
Model	(1)	(2)	(3)	(4)	(5)	(6)
$ServiceUse_{jt}$	0.019	0.013	0.024	0.017	0.010	0.006
	(0.022)	(0.012)	(0.045)	(0.023)	(0.045)	(0.012)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Customer fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Customer-specific time trends	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.721	0.726	0.778	0.781	0.784	0.789

Notes. The unit of analysis is customer (*j*) – week (*t*). The number of observations is 5,562,207. Numbers in parentheses are two-way clustered standard errors at customer and week level.

Contrary to this initial conjecture, our estimation results in Table 14 show that the coefficients of $ServiceUse_{jt}$ on all outcome measures are *not* statistically significant. This indicates that the online purchase patterns of existing customers did not change significantly as a result of using the off-line return service. Therefore, the channel shift of existing customers from online to off-line stores is not a major force that drives the enhanced store channel performance shown earlier.

Finally, we note that the customers who choose to use the return service (treated customers) might be systematically different from those who do not (control customers). For a robustness check, we conduct the coarsened exact matching as we do in Section 6.3 to refine the comparability of treated and control group customers. The estimation results are qualitatively consistent with those in Tables 13 and 14. Details of this robustness check are reported in Online Tables A13–A15.

8. Conclusions

Increasing product returns causes significant problems to online retailers. Recently, online retailers have started to collaborate with location partners (third-party retailers with physical stores), which offers a convenient store return option to customers of online retailers. To the best of our knowledge, this paper is the first that provides a rigorous analysis for the economic value of such an online–off-line return partnership to the location partner. Our analysis shows that the return partnership creates significant economic value to the location partner: the introduction of the return service in the location partner's stores increases transactions and net revenue in both online and off-line channels of the location partner.

We attribute the improved performance to two underlying effects: customer acquisition and supercharging. We find that offering the return service introduces new customers to both store and online channels of the location partner. Its effect on the online channel is particularly interesting because the store return option brings in foot traffic to off-line stores. For this effect, we offer two plausible explanations with empirical support: the enhanced brand awareness to "online-savvy" customers and the convenience of returns that reduces their perceived risk associated with online purchase. In addition to customer acquisition, the return service induces existing customers to visit stores more frequently and to purchase more items. This supercharging effect is significant in the off-line channel and can be explained by the fact that it is during a store visit that existing customers are exposed to the new mode of interaction with the retailer. However, this effect is not significant in the online channel, indicating that more off-line returns do not have a negative impact on the online sales of existing customers. This also implies that a channel shift of existing customers from the online channel to the store channel as a result of off-line returns is not a major driving force for the improved store performance. Table 15 summarizes our findings.

Table 15. Summary of the Effects of Return Partnership on Customers' Purchase Behavior

Channel	Customer acquisition effect	Customer supercharging effect
Off-line	Positive, significant	Positive, significant
Online	Positive, significant	Insignificant

Our findings underscore the ability of such unique partnerships to generate value for the stakeholders. For firms in today's competitive retail environment, it is important to understand the implications of our results as they seek to enter or expand similar collaborations; our research partner companies have been able to use the results of this study to strengthen their partnership further. The partnership helps enhance the value of the location partner's existing customers by inducing them to visit stores more frequently and to purchase more items. The success of the return partnership hinges on the preference of customers to return online purchases to stores, and our study demonstrates that the partnership is indeed successful in providing this option to online retailers; see Figure 1 on the volume of returns to off-line stores. The high cost of operating stores (or even physical showrooms) makes return partnerships very appealing to online retailers. Moreover, because customers who return more are also likely to be more valuable to the retailer in the long run (Dugdale 2010), this is a promising option for online retailers looking for a sales boost.

Although omnichannel location partners benefit in both channels, the convincing performance improvement we observe in the off-line channel alone suggests that off-line-only retailers (such as shopping malls, independent bookstores, etc.) could also explore collaborations with third-party firms such as Happy Returns to offer the return service. The fundamental value proposition of the Happy Returns business model (as demonstrated in our findings)—the conversion of online customers to revenue for the off-line store—is especially valuable for off-line stores that experience declining foot traffic because of the prevalence of online shopping. Therefore, both new and established off-line retailers may benefit from these partnerships. However, the specific magnitude of the impact can be idiosyncratic. If a location partner sells product categories similar to returned items (e.g., some product categories sold by both Amazon and Kohl's), it has more cross-selling opportunities—possibly purchasing a better fit product after inspecting it in person in the same category that the customer is returning—which benefits the location partner but possibly hurts online retailers through a substitution effect. In such cases, the extent of the impact may be larger than in the present study. Likewise, it is plausible that the returning customers may not like a location partner's products and simply drop off returns without adding any value to the location partner.

Although our paper focuses on the benefits of a return partnership, it would be interesting to examine the direct (e.g., additional labor cost, additional storage space for returned items) and indirect (e.g., longer waiting time because of some customers' return processing, reduced store assistance from employees who may be helping Happy Returns customers) costs associated with the partnership. To do so, one needs to analyze hourly labor and store traffic data, which, unfortunately, we do not have. Also, if store and website traffic data are available, it would be useful to further understand how much benefit is coming from an increase in traffic and a change in the customer population. Finally, another interesting avenue of future research is to examine the impact of a return partnership on online retailers and future customer purchases. This paper may serve as the first step to many such analyses that evaluate online–off-line retail partnerships.

Acknowledgments

The authors thank their research partners, Happy Returns and its location retail partner, for their immense support. Special thanks go to Happy Returns' Chief Executive Officer David Sobie, Caitlin Roberson, Cassie Layton, Rachel LaConti, Isabelle Darby, Elizabeth Owens, and Dayna Wafer. The authors also gratefully acknowledge helpful comments from the department editor Hau Lee; the associate editor; two anonymous reviewers; Hyun Seok Lee; Chloe Glaeser; Joseph Xu; seminar participants at Purdue University, University of British Columbia, and University of Washington; and participants at CIST, MSOM, and participants at Conference on Information Systems and Technology (CIST), Manufacturing & Service Operations (M&SOM) Annual Conference, and Workshop on Information Systems and Economics (WISE).

References

Altug MS, Aydinliyim T (2016) Counteracting strategic purchase deferrals: The impact of online retailers' return policy decisions. *Manufacturing Service Oper. Management* 18(3):376–392.

Anderson E, Fong N, Simester D, Tucker C (2010) How sales taxes affect customer and firm behavior: The role of search on the internet. *J. Marketing Res.* 47(2):229–239.

Angrist JD, Pischke JS (2009) *Mostly Harmless Econometrics: An Empiricist's Companion* (Princeton University Press, Princeton, NJ).

Armstrong P (2016) Why your business should be using Google Trends. *Forbes.* https://www.forbes.com/sites/paularmstrongtech/2016/04/01/why-your-business-should-be-using-google-trends/.

Autor DH (2003) Outsourcing at will: The contribution of unjust dismissal doctrine to the growth of employment outsourcing. *J. Labor Econom.* 21(1):1–42.

Babar Y, Burtch G (2020) Examining the heterogeneous impact of ride-hailing services on public transit use. *Inform. Systems Res.* 31(3):820–834.

Bechwati N, Siegal W (2005) The impact of the prechoice process on product returns. *J. Marketing Res.* 42(3):358–367.

Bell D, Gallino S, Moreno A (2018) Offline showrooms in omnichannel retail: Demand and operational benefits. *Management Sci.* 64(4):1629–1651.

Bell D, Gallino S, Moreno A (2020) Customer supercharging in experience-centric channels. *Management Sci.* 66(9):4096–4107.

Bertrand M, Duflo E, Mullainathan S (2004) How much should we trust differences-in-differences estimates? *Quart. J. Econom.* 119(1):249–275.

Burtch G, Carnahan S, Greenwood BN (2018) Can you gig it? An empirical examination of the gig economy and entrepreneurial activity. *Management Sci.* 64(12):5497–5520.

Chen B, Chen J (2017) When to introduce an online channel, and offer money back guarantees and personalized pricing? *Eur. J. Oper. Res.* 257(2):614–624.

Choi H, Varian H (2012) Predicting the present with Google Trends. *Econom. Rec.* 88(S1):2–9.

Danziger P (2019) Retail downsizing will accelerate, with UBS predicting 75,000 stores will be forced to close by 2026. *Forbes.* https://www.forbes.com/sites/pamdanziger/2019/04/10/retail-downsizing-will-accelerate-as-75000-stores-will-be-forced-to-close-by-2026/.

Donald SG, Lang K (2007) Inference with difference-in-differences and other panel data. *Rev. Econom. Statist.* 89(2):221–233.

Dugdale A (2010) Zappos' best customers are also the ones who return the most orders. *Fast Company Online* (April 13), https://www.fastcompany.com/1614648/zappos-best-customers-are-also-ones-who-return-most-orders.

Dutt N, King AA (2014) The judgment of garbage: End-of-pipe treatment and waste reduction. *Management Sci.* 60(7):1812–1828.

Ertekin N, Agrawal A (2021) How does a return period policy change affect multichannel retailer profitability? *Manufacturing Service Oper. Management* 23(1):210–229.

Forde M (2019) FedEx partners with Walgreens on returns in preparation for peak season. Supply Chain Dive (October 8), https://www.supplychaindive.com/news/fedex-partners-with-walgreens-on-returns/564604/.

Gallino S, Moreno A (2014) Integration of online and offline channels in retail: The impact of sharing reliable inventory availability information. *Management Sci.* 60(6):1434–1451.

Gao F, Su X (2017a) Omnichannel retail operations with buy-online-and-pick-up-in-store. *Management Sci.* 63(8):2478–2492.

Gao F, Su X (2017b) Online and offline information for omnichannel retailing. *Manufacturing Service Oper. Management* 19(1):84–98.

Goodman-Bacon A (2018) Difference-in-differences with variation in treatment timing. NBER Working Paper No. 25018, National Bureau of Economic Research, Cambridge, MA.

Greenwood BN, Wattal S (2017) Show me the way to go home: An empirical investigation of ride-sharing and alcohol related motor vehicle fatalities. *Management Inform. Systems Quart.* 41(1):163–187.

Happy Returns (2020) Accessed March 1, 2020, https://www.happyreturns.com/.

Iacus SM, King G, Porro G (2009) CEM: Software for coarsened exact matching. *J. Statist. Software* 30(9):1–27.

Iacus SM, King G, Porro G (2012) Causal inference without balance checking: Coarsened exact matching. *Political Anal.* 20(1):1–24.

Iacus SM, King G, Porro G (2018) Package "cem." Accessed June 24, 2020, https://CRAN.R-project.org/package=cem.

Kapner S (2021) Covid-19 rewrote the rules of shopping. What is next? *Wall Street Journal* (March 12). Accessed October 13, 2021, https://www.wsj.com/articles/covid-19-rewrote-the-rules-of-shopping-what-is-next-11615561232.

Kohl's (2019) Kohl's announces expansion of Amazon returns to all Kohl's stores nationwide. Accessed October 13, 2020, https://corporate.kohls.com/news/archive-/2019/April/kohl-s-announces-expansion-of-amazon-returns-to-all-kohl-s-store/.

Kumar A, Mehra A, Kumar S (2019) Why do stores drive online sales? Evidence of underlying mechanisms from a multichannel retailer. *Inform. Systems Res.* 30(1):319–338.

Lam S, Vandenbosch M, Hulland J, Pearce M (2001) Evaluating promotions in shopping environments: Decomposing sales response into attraction, conversion, and spending effects. *Marketing Sci.* 20(2):194–215.

Lee H, Kesavan S, Deshpande V (2020) Managing the impact of fitting-room traffic on retail sales: Using labor to reduce phantom stockouts. *Manufacturing Service Oper. Management* Forthcoming.

Mulpuru S (2017) Think tank: The opportunity in online returns for apparel retailers. *WWD Online* (July 6). Accessed October 13, 2020, https://wwd.com/business-news/business-features/sucharita-mulpuru-think-tank-returns-10939339/.

Nageswaran L, Cho S-H, Scheller-Wolf A (2020) Consumer return policies in omnichannel operations. *Management Sci.* 66(12):5558–5575.

Nageswaran L, Hwang EH, Cho S-H (2021) Offline returns for online retailers via partnership. Working paper, University of Washington, Seattle.

Narvar (2019) The state of online returns: A global study. Accessed May 3, 2021, https://see.narvar.com/Consumer_Report-Returns_LP.html.

Nielsen (2021) DMA regions. Accessed May 19, 2021, https://www.nielsen.com/us/en/contact-us/intl-campaigns/dma-maps/.

Ofek E, Katona Z, Sarvary M (2011) "Bricks and clicks": The impact of product returns on the strategies of multichannel retailers. *Marketing Sci.* 30(1):42–60.

Perdikaki O, Kesavan S, Swaminathan J (2012) Effect of traffic on sales and conversion rates of retail stores. *Manufacturing Service Oper. Management* 14(1):145–162.

Peterson H (2019) Kohl's CEO said Amazon returns boosted traffic and data shows the jump could be as high as 24%. *Business Insider* (August 20). Accessed October 13, 2021, https://www.businessinsider.com.au/amazon-returns-help-boost-kohls-stores-shopper-traffic-2019-8.

Reagan C (2019) That sweater you don't like is a trillion-dollar problem for retailers. These companies want to fix it. *CNBC* (January 12). https://www.cnbc.com/2019/01/10/growing-online-sales-means-more-returns-and-trash-for-landfills.html.

Risley J (2018) Happy Returns makes major retail expansion with Paper Source partnership. Accessed May 3, 2021, https://www.digitalcommerce360.com/2018/08/29/happy-returns-makes-major-retail-expansion-with-paper-source-partnership/.

Rosenbaum PR, Rubin DB (1983) The central role of the propensity score in observational studies for causal effects. *Biometrika* 70(1):41–55.

Shang G, Ferguson ME, Galbreth MR (2019) Where should I focus my return reduction efforts? Empirical guidance for retailers. *Decision Sci.* 50(4):877–909.

Shulman J, Cunha M Jr, Saint Clair J (2015) Consumer uncertainty and purchase decision reversals: Theory and evidence. *Marketing Sci.* 34(4):590–605.

Su X (2009) Consumer returns policies and supply chain performance. *Manufacturing Service Oper. Management* 11(4):595–612.

Ulku M, Dailey L, Yayla-Kullu H (2013) Serving fraudulent consumers? The impact of return policies on retailer's profitability. *Service Sci.* 5(4):296–309.

Valinsky J (2020) Macy's is closing 28 stores and a Bloomingdale's store. *CNN* (January 8). https://williamsonsource.com/macys-announces-store-closures/.

Wang K, Goldfarb A (2017) Can offline stores drive online sales? *J. Marketing Res.* 54(5):706–719.

Wang G, Zheng R, Dai T (2021) Does transportation mean transplantation? Impact of new airline routes on sharing of cadaveric kidneys. *Management Sci.* Forthcoming.

Wood S (2001) Remote purchase environments: The influence of return policy leniency on two-stage decision processes. *J. Marketing Res.* 38(2):157–169.

Wooldridge JM (2010) *Econometric Analysis of Cross Section and Panel Data* (MIT Press, Boston).